

PARALLEL PROCESSING AND FLYNN'S CLASSIFICATION

1. What is the main advantage of parallel processing?

- a) Reduces cost
- b) Increases execution speed**
- c) Reduces power consumption
- d) Increases memory size

2. Which of the following architectures is NOT a part of Flynn's classification?

- a) SISD
- b) SIMD
- c) MIMD
- d) SMP**

3. In SIMD architecture, what is common among all processing elements?

- a) Instruction**
- b) Data
- c) Execution time
- d) Registers

4. Which of the following is an example of an MIMD system?

- a) Single-core processor
- b) Multi-core processor**
- c) GPU
- d) Vector processor

5. Which of the following is an example of SIMD processing?

- a) Multithreading
- b) Vector processing**
- c) Pipelining
- d) Distributed computing

6. Which unit in a parallel system is responsible for controlling execution flow?

- a) Arithmetic Logic Unit
- b) Control Unit**
- c) Cache Memory
- d) Floating Point Unit

7. Which parallel processing model is least commonly implemented in real-world systems?

- a) SISD
- b) SIMD
- c) MISD
- d) MIMD

8. What is the role of a scheduler in parallel processing?

- a) Assign tasks to processors
- b) Increase CPU power
- c) Manage network connections
- d) Improve memory usage

9. what is the data stream in computer architecture?

- a) Data traffic exchange between memory and processing unit.
- b) Data traffic exchange between memory and CPU
- c) Data stream between processor and memory.
- d) Sequence of instructions performed by the processing unit.

10. In which scenario is parallel processing NOT useful?

- a) Large-scale scientific simulations
- b) Gaming
- c) Simple arithmetic calculations
- d) Image processing

11. Flynn's classification is primarily based on which of the following factors?

- a) Number of CPU Cores and memory architecture
- b) Number of control units and clock speed
- c) Number of instruction streams and data streams
- d) Number of registers and cache hierarchy

12. Why SISD is not used in parallel processing?

- a) It executes instructions sequentially
- b) It carry out simultaneous data-processing
- c) It carry out multiple instructions on multiple data sets
- d) None

13. what is the instruction stream in computer architecture?

- a)Data traffic exchange between memory and processing unit.
- b)Data traffic exchange between memory and CPU
- c)Data stream between processor and memory.
- d)Sequence of instructions performed by the processing unit.**

14. What type of computing system is an SISD system?

- a)Uniprocessor machine executing a single instruction.**
- b)Multiprocessor machine executing a multiple instruction.
- c)Multiprocessor machine executing a different instruction.
- d)Uniprocessor machine executing a different instruction.

15. What is the memory configuration in Shared MIMD?

- a) PU's have connection with Global memory.**
- b) PU's have shared cache memory.
- c) PU's have individual local memory.
- d) PU's have no memory access.